

Matrix Algebra

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Today's class

- **Vectors**

- Vector addition and scalar multiplication
- Vector norms
- Dot products
 - Angle between two vectors
- Projections 1

- **Matrices**

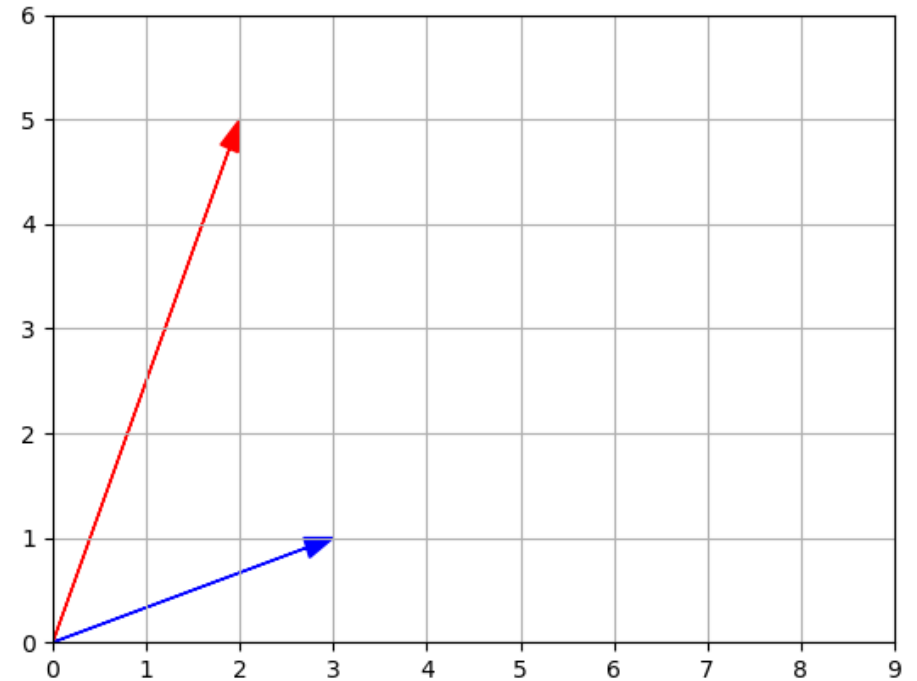
- Matrix addition, multiplication, transpose, and inversion
- Projection matrices
- Eigenvalues and eigenvectors
- Projections 2
- Matrix polynomials

Resources

- Aurelien Geron's brief but excellent summary of matrix algebra:
https://github.com/ageron/handson-ml3/blob/main/math_linear_algebra.ipynb
- Richard Bronson, "*Schaum's Outline of Matrix Operations*." McGraw-Hill, 2011.
 - This book has plenty of **solved problems** along with theory.
- Deisenroth, Marc *et al.* (2020). "*Mathematics for machine learning*." Cambridge University Press.
 - Chapter 3: 3.1, 3.2, 3.3, 3.4, 3.9,
 - Chapter 4: 4.2, 4.4 (optional), 4.5 (optional).

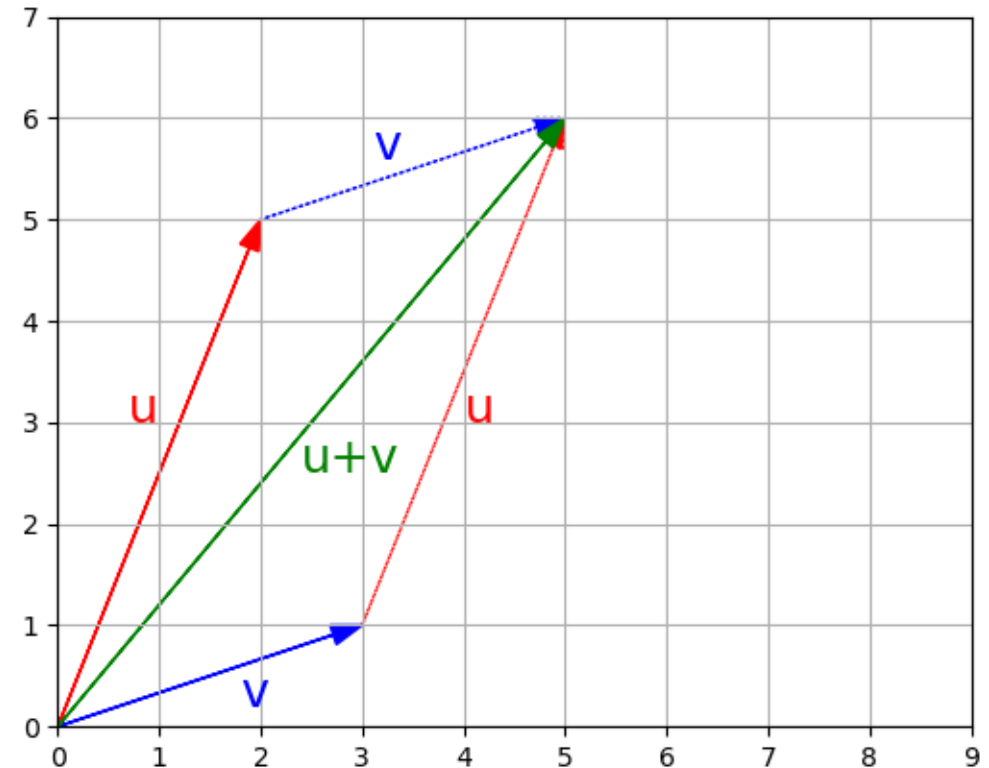
Vectors

- A vector is a quantity defined by
 - a magnitude and
 - a direction.
- Given a vector $\mathbf{u} = [u_1, \dots, u_n]$,
 - The $u_i, 1 \leq i \leq n$ are called its coordinates.
 - This vector resides in an n -dimensional space.



Vectors

- Vector addition:
 - Given two vectors $\mathbf{u} = [u_1, \dots, u_n]$ and $\mathbf{v} = [v_1, \dots, v_n]$,
 - $\mathbf{u} + \mathbf{v} = [u_1 + v_1, \dots, u_n + v_n]$
 - In other words, coordinate wise addition



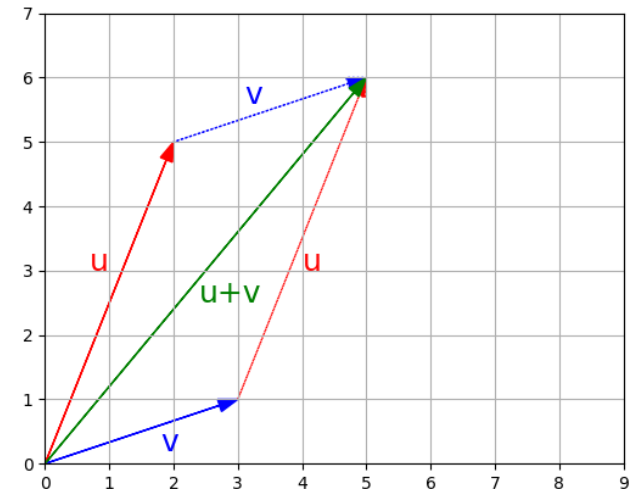
Vectors

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- **Properties**

- Vector addition is commutative, i.e., $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- It is also associative, i.e., $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$



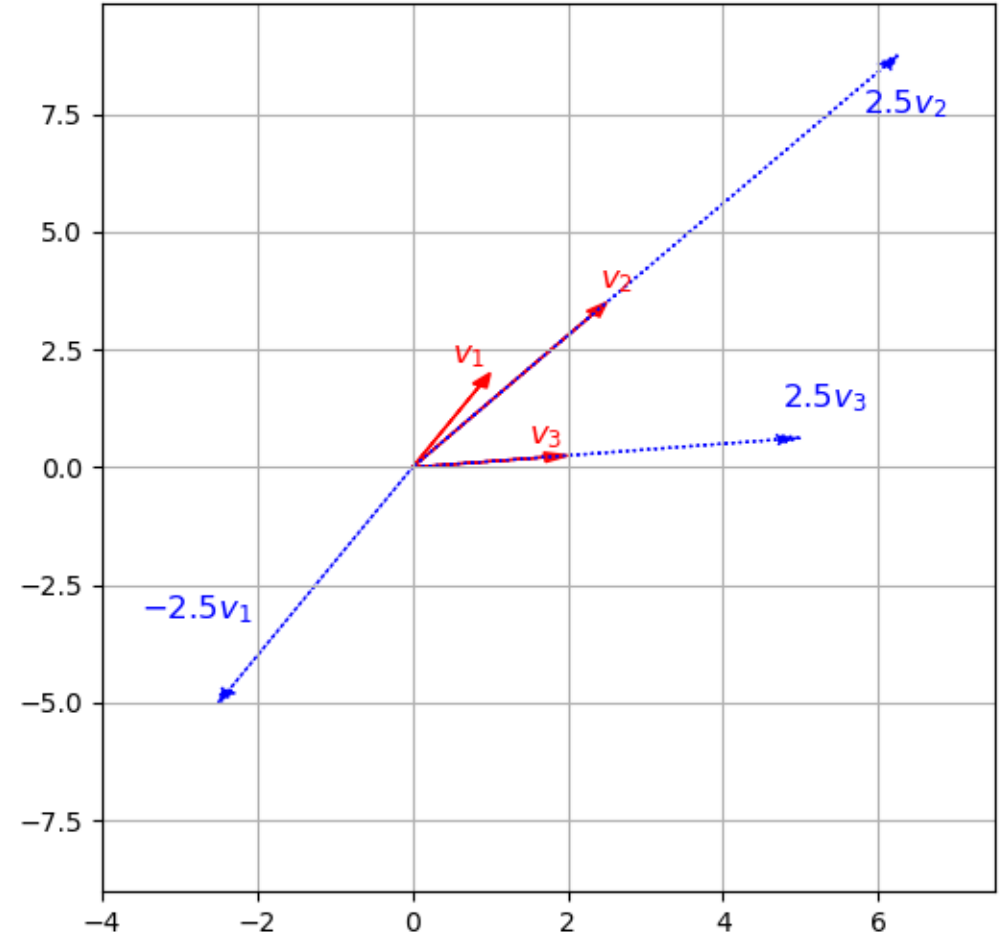
Vectors

- **Scalar multiplication:**

- Given a vectors $\mathbf{u} = [u_1, \dots, u_n]$ and a scalar λ ,
- $\lambda \mathbf{u} = [\lambda u_1, \dots, \lambda u_n]$
- In other words, coordinatewise multiplication

- Along the same lines,

- $\frac{1}{\lambda} \mathbf{u} = \left[\frac{1}{\lambda} u_1, \dots, \frac{1}{\lambda} u_n \right]$



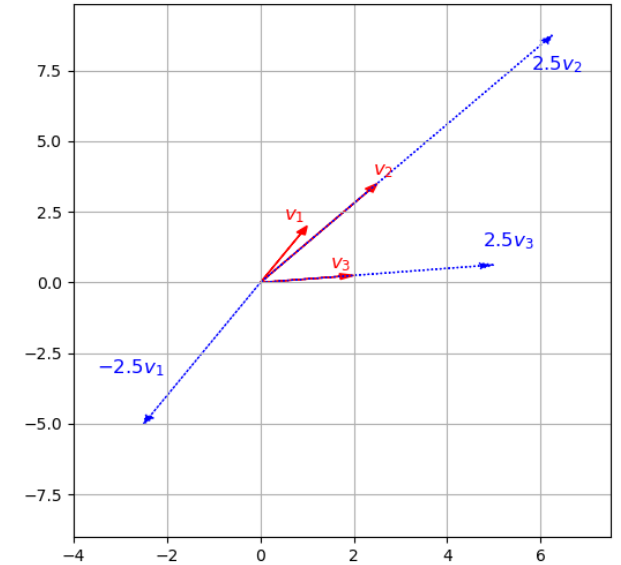
Vectors

- Scalar multiplication:

- Given a vectors $\mathbf{u} = [u_1, \dots, u_n]$ and a scalar λ ,
- $\lambda \mathbf{u} = [\lambda u_1, \dots, \lambda u_n]$

- **Properties**

- Commutative, i.e., $\lambda \mathbf{u} = \mathbf{u} \lambda$
- Associative, i.e., $\lambda_1(\lambda_2 \mathbf{u}) = (\lambda_1 \lambda_2) \mathbf{u}$
- Distributive over vector addition, i.e., $\lambda(\mathbf{u} + \mathbf{v}) = (\lambda \mathbf{u}) + (\lambda \mathbf{v})$



Vector Norms

- The **norm** of a vector $\mathbf{u}$, noted $\|\mathbf{u}\|$, is a measure of the length (a.k.a. the magnitude) of $\mathbf{u}$.
 - Recall our notation, $\mathbf{u} = [u_1, \dots, u_n]$.
- Euclidean norm or the $\mathcal{L}^2$ -norm is defined as
 - $\|\mathbf{u}\| = \|\mathbf{u}\|_2 = \sqrt{\sum_{i=1}^n u_i^2}$
- The $\mathcal{L}^1$ -norm is defined as
 - $\|\mathbf{u}\|_1 = \sum_{i=1}^n |u_i|$

Vector Norms

- $\mathcal{L}^2$ -norm is defined as

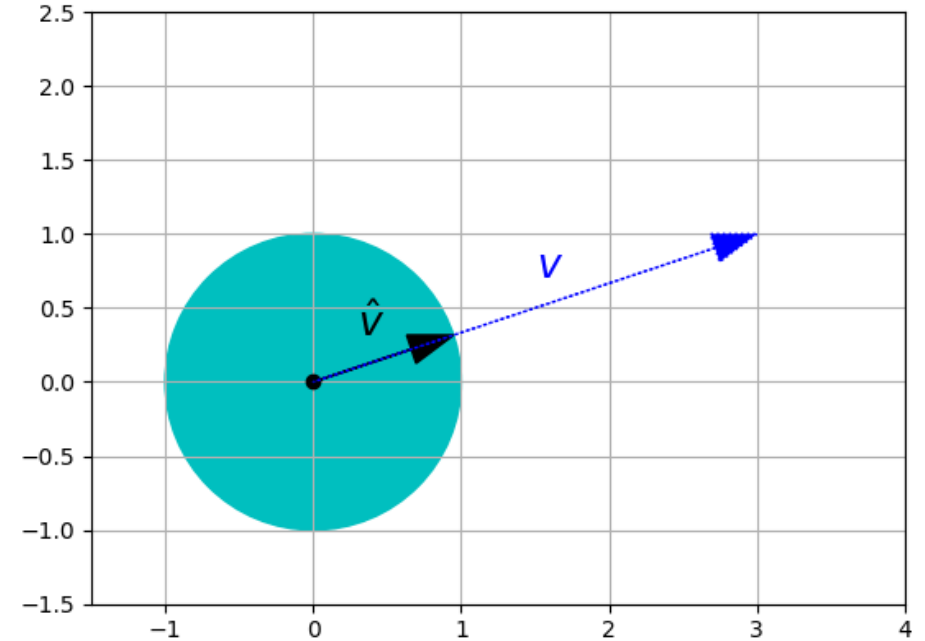
- $\|\mathbf{u}\| = \sqrt{\sum_{i=1}^n u_i^2}$

- A unit vector is one whose norm = 1
- Normalization:

- The normalized vector of $\mathbf{u}$,

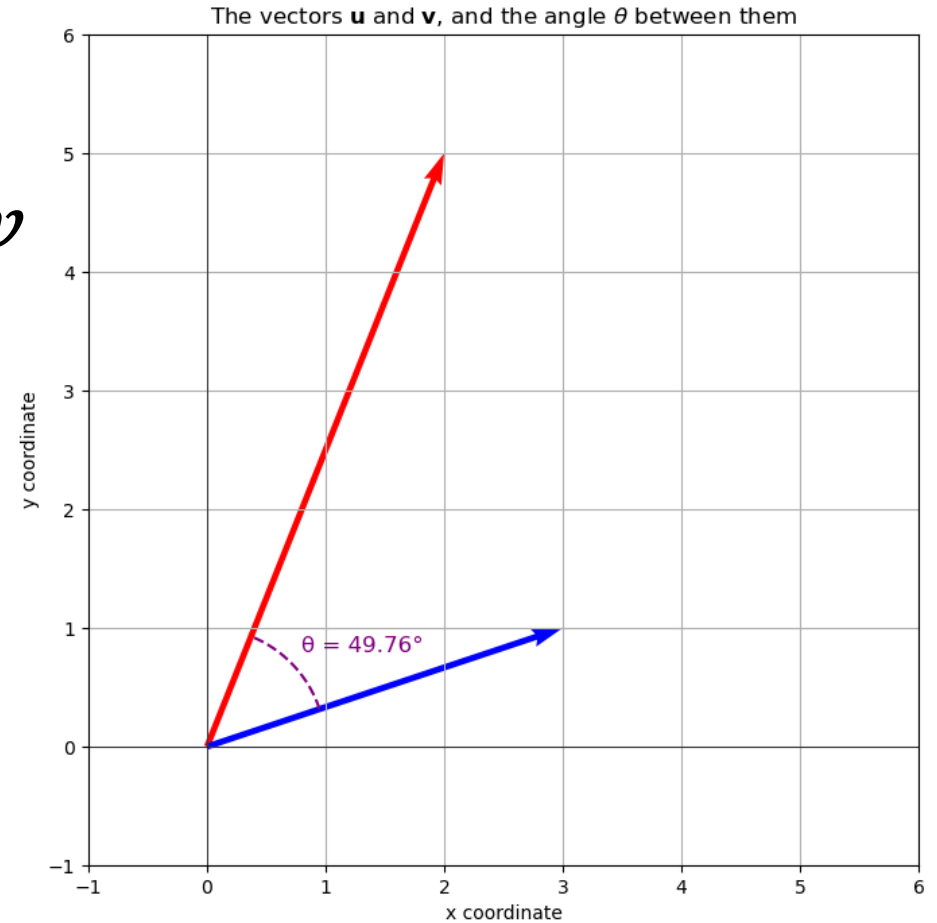
$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

- Same direction as $\mathbf{u}$
 - Unit vector



Inner Products

- The inner product between vectors $\mathbf{u}$ and $\mathbf{v}$ is given by
 - $\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i \times v_i$
 - Also denoted by $\langle \mathbf{u}, \mathbf{v} \rangle$
- Inner products are also called “**Dot Products**”
- Measure of the “angle” between the two vectors
 - $\theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \times \|\mathbf{v}\|} \right)$



Projections 1: on to an axis

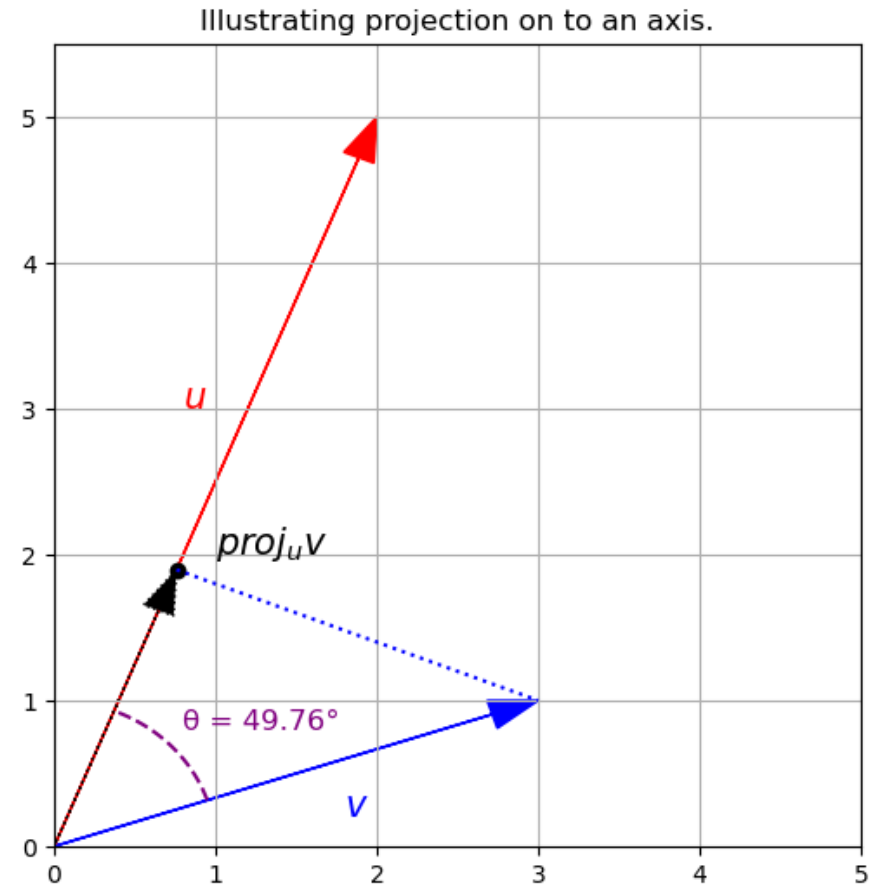
Given vectors $\mathbf{u}$ and $\mathbf{v}$, the “**projection**” of $\mathbf{v}$ on to the axis along $\mathbf{u}$ is given by

$$\text{proj}_{\mathbf{u}} \mathbf{v} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|^2} \right) \mathbf{u}$$

- This is also given by

$$\text{proj}_{\mathbf{u}} \mathbf{v} = (\mathbf{v} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}$$

- Notice: $\mathbf{v} - \text{proj}_{\mathbf{u}} \mathbf{v}$ is perpendicular to $\mathbf{u}$



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Matrices

- A matrix is a rectangular array of scalars arranged in rows and columns, for example:

$$\begin{bmatrix} 10 & -33.3 \\ -50.0 & 0 \end{bmatrix}$$

- Matrices are convenient and very efficient to **run operations on many vectors at a time.**
- We will also see that they are great at representing and performing linear transformations such as
 - rotations,
 - translations,
 - scaling, and
 - **projections**

Matrices

- **Size:** The size of a matrix is defined by its number of rows and number of columns.

- *(rows $\times$ columns)*

- So, a (2×3) matrix has 2 rows and 3 columns.

$$\begin{bmatrix} 10 & 40 & 30 \\ 20 & 50 & 40 \end{bmatrix}_{2 \times 3}$$

- $A_{i,j}$: the entry in the i^{th} row and j^{th} column of matrix A
- Will also see: Square, triangular, diagonal, and identity matrices.

Matrices: sums and products

- Two matrices P and Q of the same size $(m \times n)$,
- Their **sum** $S = P + Q$ is defined by $S_{ij} = P_{ij} + Q_{ij}$
 - Like with vectors, matrix addition is *elementwise*.
- Matrix **products**: $P_{m \times n} \times Q_{n \times l}$
 - Defined only if $n = k$
- $S = P \times Q$
 - $S_{ij} = \sum_{t=1}^n P_{it} \times Q_{tl}$
- **Scalar multiplication**: given a matrix $A_{(m \times n)}$ and a scalar λ
 - $\lambda A = [\lambda A_{ij}]_{(m \times n)}$

Matrix Transpose

- The “*transpose*” of an $(m \times n)$ matrix A is the $(n \times m)$ matrix A^T
 $(A^T)_{ij} = A_{ji}$
 - rows and columns are switched

- $A = \begin{bmatrix} 1 & 2 & 10 \\ -4 & 3 & -5 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & -4 \\ 2 & 3 \\ 10 & -5 \end{bmatrix}$

- Remember: $(A \times B \times C)^T = C^T \times B^T \times A^T$
- Symmetric matrices: $A^T = A$
- Antisymmetric matrices: $A^T = -A$

Special matrices

- Square
- Identity ($I_{n \times n}$)
- Zero ($0_{n \times n}$)
- Upper triangular
- Lower triangular
- Diagonal

Matrix Inversion

- The inverse of a (square) matrix A is another matrix B that satisfies

$$A \times B = B \times A = I$$

- Usually denoted A^{-1}
- Remember: $(M \times N)^{-1} = N^{-1} \times M^{-1}$
-  Need not necessarily exist!
 - Conditions for existence: coming soon.

Matrix vector products

- Multiplying an $(m \times n)$ matrix A , with an $(n \times 1)$ vector $\mathbf{x}$ gives an $(m \times 1)$ vector $\mathbf{b}$,

$$\mathbf{b} = A\mathbf{x}$$

- Matrix-vector products are *linear*

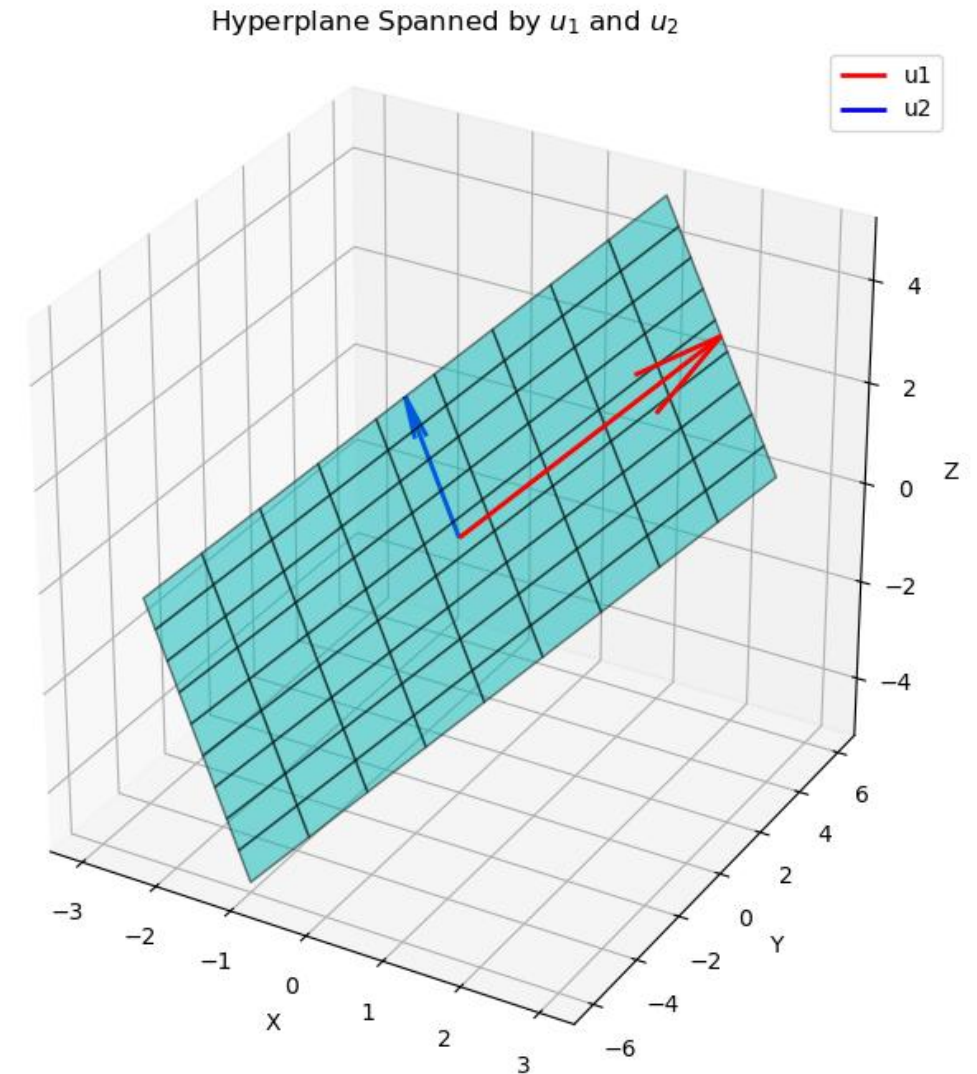
$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$$

- Also,

$$A(\lambda_1\mathbf{x} + \lambda_2\mathbf{y}) = A(\lambda_1\mathbf{x}) + A(\lambda_2\mathbf{y}) = \lambda_1 A\mathbf{x} + \lambda_2 A\mathbf{y}$$

Projections 2: on to a hyperplane

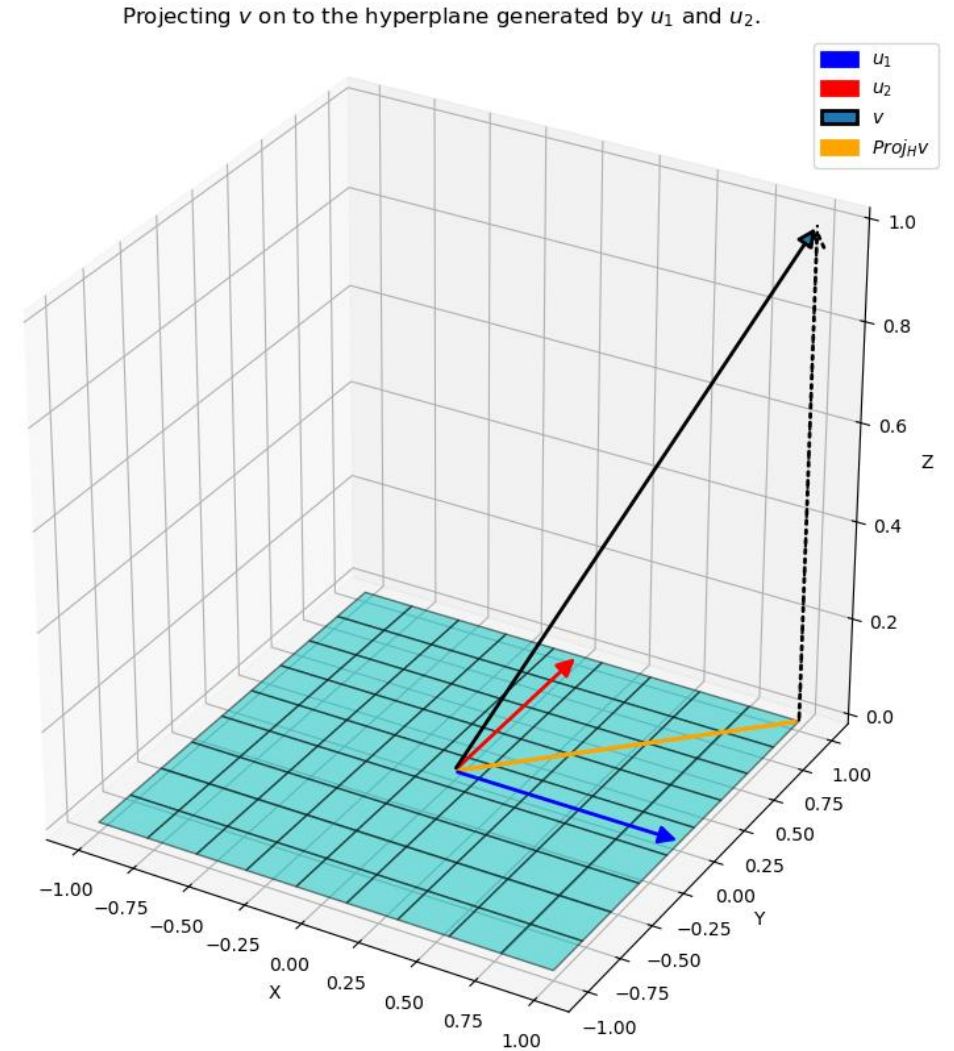
- This is what happens in many regression algorithms.
- Figure shows the hyperplane generated by $\mathbf{u}_1 = [2, 5, 3]$ and $\mathbf{u}_2 = [-1, 1, 2]$
- $\mathcal{H} = \{\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2\}$
 - λ_1, λ_2 are both real numbers.
- Express $\mathcal{H}$ in matrix-vector format?



Projections 2

- $\mathcal{H}$: hyperplane generated by $\mathbf{u}_1$ and $\mathbf{u}_2$.
- $\text{Proj}_{\mathcal{H}}\mathbf{v}$: the projection of $\mathbf{v}$ on to $\mathcal{H}$.
- Notice that $\mathbf{v} - \text{Proj}_{\mathcal{H}}\mathbf{v}$ is perpendicular to $\mathcal{H}$.
- $U = [\mathbf{u}_1 \ \mathbf{u}_2]$ (has $\mathbf{u}_1$ and $\mathbf{u}_2$ as columns)

$$\text{Proj}_{\mathcal{H}}\mathbf{v} = U(U^T U)^{-1}U^T \mathbf{v}$$



Eigenvalues and Eigenvectors

- An **eigenvector** of a square matrix M (also called a **characteristic vector**) is a non-zero vector that **remains on the same line/axis** after multiplication by M .

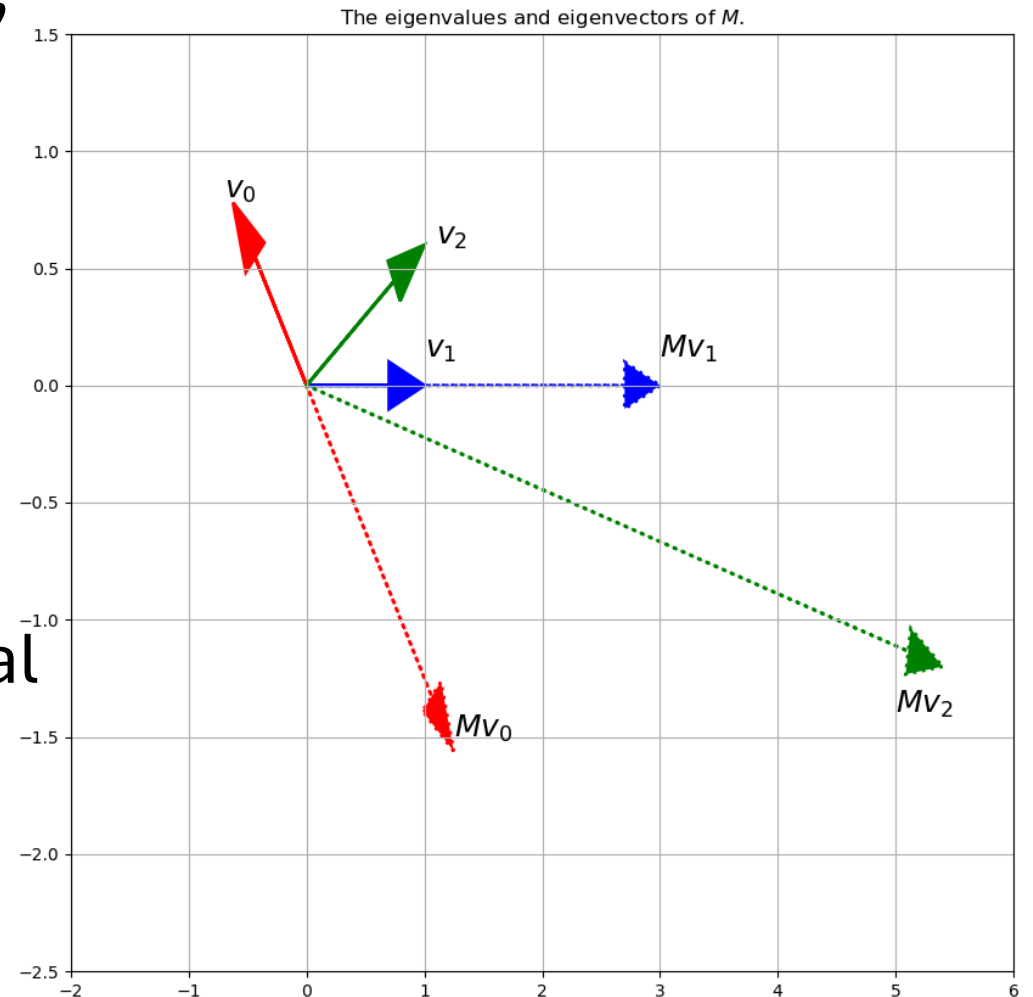
$$M\boldsymbol{v} = \lambda\boldsymbol{v}$$

- $\boldsymbol{v}$: an **eigenvector** of M
- λ : the **eigenvalue** associated with $\boldsymbol{v}$.
- Invertibility $\Leftrightarrow$ **no zero eigenvalues**

Eigenvalues and Eigenvectors

$$Mv = \lambda v$$

- v : an **eigenvector** of M
- λ : the **eigenvalue** associated with v .
- Fun fact: the eigenvalues of any upper or lower triangular matrix **are** the diagonal entries.

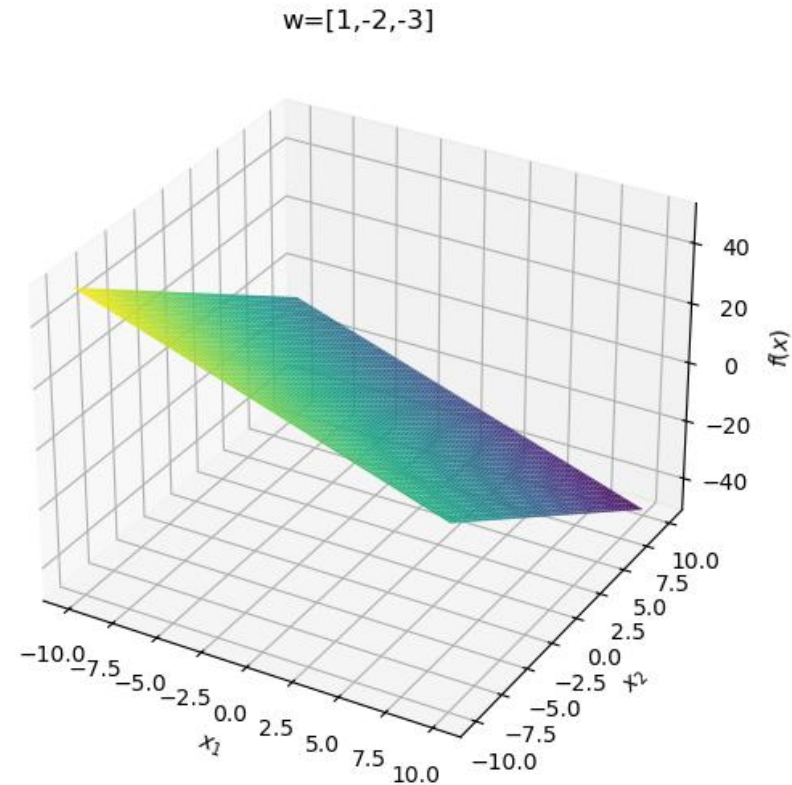
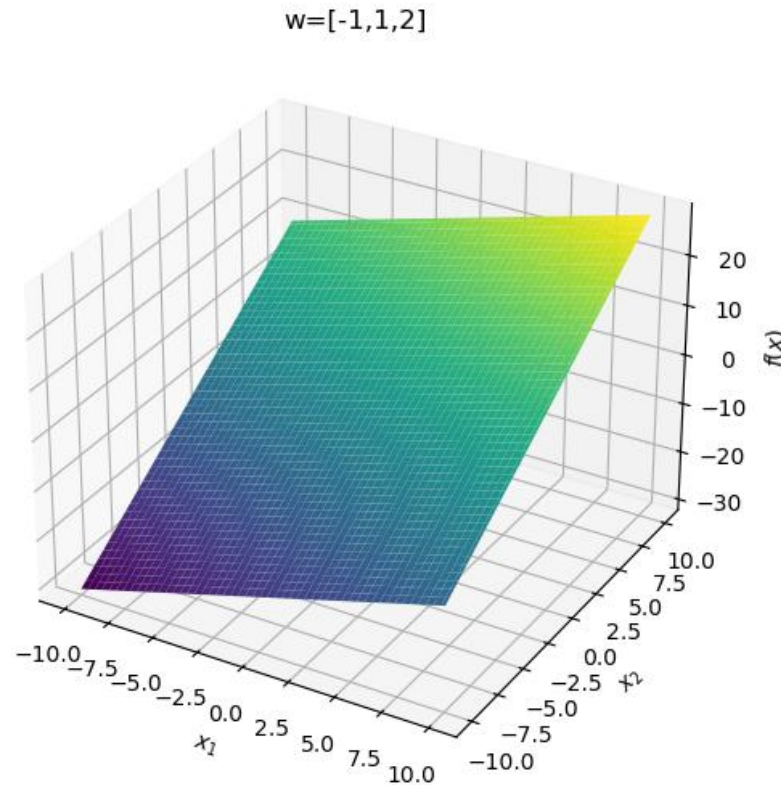


Matrix Polynomials

- Linear and quadratic functions *of vectors*.
- Why? Expressing predictors and loss functions as matrix-vector products
 - Viewpoint: extremely useful for deep learning.
- $f(x_1, x_2) = w_0 + w_1x_1 + w_2x_2$
 - Function of two variables x_1, x_2
 - Parameterized by 3 coefficients: w_0, w_1, w_2
- $f(\mathbf{x}) = w_0 + \mathbf{w}^T \mathbf{x}$
 - Linear

Matrix Polynomials

- $f(\mathbf{x}) = w_0 + \mathbf{w} \cdot \mathbf{x}$
- What does this function look like?



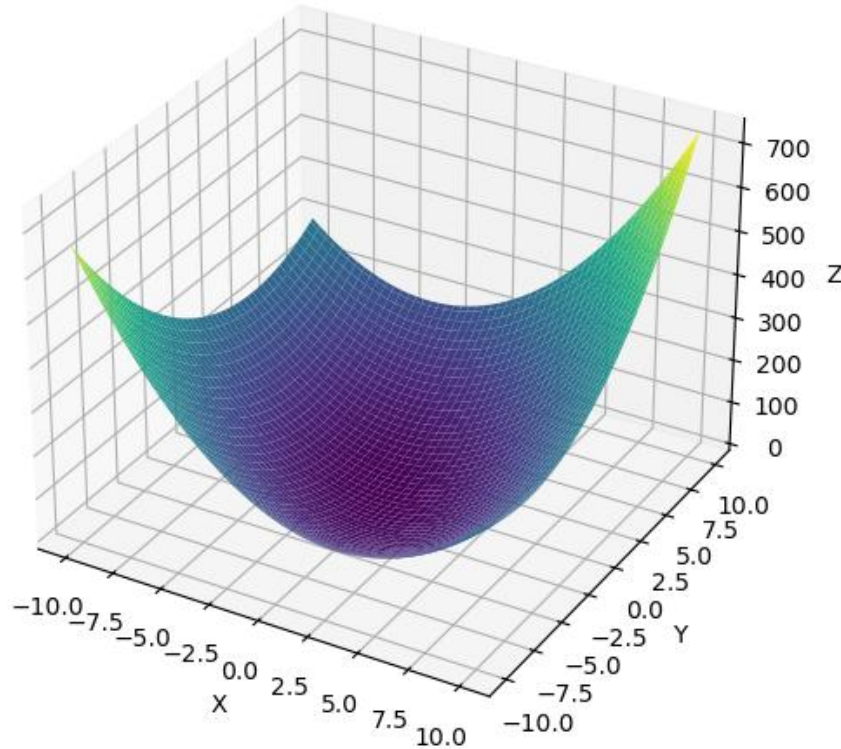
Matrix Polynomials

- Quadratic functions
- $f(x_1, x_2) = w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_1x_2 + w_5x_2^2$
 - Function of two variables x_1, x_2
 - Parameterized by 6 coefficients: $w_0, \dots, w_5$
- General form: $f(\mathbf{x}) = w_0 + \mathbf{x}^T W_1 \mathbf{x} + \mathbf{w}_2^T \mathbf{x}$
 - Similar to the **scalar** quadratic forms we're used to
 $ax^2 + bx + c$
 - Shape: when $a > 0$ and when $a < 0$?
- Alternate popular notation
$$f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$$

Matrix Polynomials

- Effect of eigenvalues

All eigenvalues positive.



One positive and one negative eigenvalue.

